

SUMMARY

I'm a **Design Engineer** with over **three years of experience** in the **aerospace and aviation industry**. My passion lies in designing and optimizing **aerostructures** for both commercial and defense applications. I love using tools like **CATIA, SolidWorks, and ANSYS** to bring ideas to life, making products **lighter, stronger, and more efficient**. Over the years, I've helped companies **reduce weight by up to 10%** and **streamline design processes by 25%**. Beyond design, I enjoy working with **cross-functional teams**, solving complex challenges, and ensuring everything we create meets the highest **FAA/EASA standards**.

SKILLS

Design & Engineering: SolidWorks, AutoCAD, CATIA, CREO, Siemens NX, PTC Creo, ANSYS, NASTRAN, ABAQUS, Hypermesh, Material Selection & Testing.

Efficiency & Process Optimization: Lean Manufacturing, SAP ERP, Cycle Time Reduction, FAA/EASA Compliance, RCCA, 5S Principles, Process Standardization & Defect Reduction, Design for Manufacturability (DFM) & Design for Assembly (DFA), Failure Modes and Effects Analysis (FMEA)

Data & Programming Skills: Python, MATLAB, R, Power BI, Simulink, Microsoft Project, Primavera P6 (Task Prioritization, Cost Analysis, Budgeting), Artificial Intelligence & Machine Learning for Engineering Optimization

Tooling & Equipment Design: Injection Molding, Welding Equipment, Product Lifecycle Management, Additive Manufacturing (3D Printing) & Rapid Prototyping, Robotics & Automation for Manufacturing

WORK EXPERIENCE

Quality engineer | Spirit AeroSystems | USA | July 2024 – Present

- Designed and optimized **aerospace structures** for **Boeing commercial** while ensuring **FAA/EASA compliance**.
- Created **3D models** and technical drawings using **CATIA V5**, enhancing design accuracy and manufacturability.
- Used **FEA tools (ANSYS, NASTRAN, Hypermesh)** to conduct **structural integrity tests** and optimize material usage.
- Identified **root causes of design and process issues** using **RCCA techniques**, leading to better manufacturing outcomes.
- Worked closely with **Boeing and Spirit AeroSystems teams** to align on product requirements.
- Explored **new composite materials** to improve **performance and cost-efficiency**.

Design Engineer | Adaptive Aerospace Group | USA | Jan 2023 – June 2024

- Designed **aircraft structures and control systems**, reducing **design time by 25%**.
- Worked on **Unmanned Aircraft Systems (UAS)**, increasing **operational safety by 30%** using **MATLAB & Simulink**.
- Led a **cross-functional team** to develop **AI-powered pilot response models**, improving **system accuracy by 20%**.
- Automated **flight dynamics testing**, cutting **prototype evaluation time by 40%** using **Python**.
- Ensured **projects were delivered on time and met all FAA compliance requirements**.

Design Engineer | Cyient | Hyderabad, India | June 2021 – May 2022

- Improved **manufacturing efficiency by 20%** and reduced **aircraft weight by 2%** through **smart engineering solutions**.
- Led **cost-reduction initiatives**, saving up to **20% on production expenses**.
- Created **detailed repair manuals**, helping clients cut costs by **21%**.
- Worked closely with clients and teams to **ensure seamless project execution**.

Design Engineer Intern | Cyient | Hyderabad, India | March 2019 – May 2019

- Assisted in **designing 3D models and validating simulations**.
- Conducted **stress, thermal, and vibration tests** to ensure industry compliance.

EDUCATION

Master's in Mechanical Engineering

Arizona State University

May 2024

Tempe, AZ

Bachelor's in Mechanical Engineering

Amrita University

May 2021

India

PROJECTS & RESEARCH

Robotic Motion Optimization

- Developed a **Particle Swarm Optimization (PSO) algorithm**, improving motion accuracy by **15%** and speeding up computations by **30%**.

Drone Design Improvements

- Enhanced **fuel storage efficiency**, cutting **maintenance frequency by 10%** and reducing **drone size by 6%**.

Thermal Cooling System for Electronics

- Led a **six-person team** to develop an **efficient single-phase immersion cooling system** using **Autodesk Inventor**.

Finite Element Analysis (FEA) for Structural Strength

- Increased **load capacity, tensile strength, and fatigue resistance** using **advanced FEA techniques**.

Metallic Composite Wing Box Design

- Created **thin-walled box beams** in **ABAQUS**, optimizing **strength-to-weight ratio** for aircraft components.

Production Layout Optimization

- Helped a company **increase profits by 4%** (~\$2M savings) through **smart production planning techniques**.
- Reduced **labor costs by 15%** by applying **linear programming (AMPL) and efficiency methods**.

CERTIFICATIONS

- Six Sigma Green Belt** (Process Improvement & Quality Management)
- GD&T Level 1 & 2** (Precision in Engineering Design)
- Advanced 3D Modeling** (Expertise in CAD software)